sanas South African National Accreditation System

NON-CONFORMANCE, CORRECTIVE ACTION AND CLEARANCE REPORT

SANAS No.: **T1195**

Organisation Name: **Azuresky Laboratory**

NC Report Number: **AB23**

Area/Scope Assessed: **Management Requirements**

Assessor / Inspector / Expert: **A. Bougart**

PART 1: DETAILED OBSERVATION

At the time of the assessment, it was observed that Procedure ASDP001 point 2.3 states that checklist ASMF 6.3.1 will be used to actively monitor, measure and manage the performance of the company's supply. But at the time of the assessment, there was no evidence that the lab's external service providers were evaluated.

Signed (Representative):

Signed (Team Leader / Inspector): A. Bougart

MAJOR: [X] MINOR: [ ]

Reference No. of relevant Guide / Standard / Requirement: **ISO/IEC 17025 requirement 6.6.2b**

Date: **08/04/2026**

PART 2: ROOT CAUSE ANALYSIS

(Provide an analysis of the extent and cause of the nonconformity)

Personnel used previous institutional engagements to appoint external service providers & did not formalize the process of assessing & evaluating of external service providers, as stated by ASDP001 which resulted in an oversight & non-adherence to procedure

PART 3: CORRECTIVE ACTION TAKEN

(Clearly indicate what corrective action was taken, and attach supportig evidence of implementation)

- Revision of ASDP001 & equipment procedure
- Backtracking of assessment & evaluation of current service providers using a defined acceptance criteria and defined requirements using a new introduced checklist ASMF 6.6.3
- Communication of changes to personnel / One checklist has been attached as evidence 6.6.3

Date: 22/05/2026

Signed (Representative): [signature]

PART 4: CORRECTIVE ACTION VERIFIED AND CLEARED

(Comment on whether the corrective action is sufficient and appropriate)

Date:

Signed: Assessor / Inspector/Expert:

PART 5: FOLLOW-UP AT NEXT ASSESSMENT

(Comment on the implementation of the corrective actions)

Date:

Signed: Assessor / Inspector/Expert:

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AS004MF 07/25

BRAINSTORMING FORM

IRQ no: AB23

Brainstorming team

Tshepo Maponya
Keitumetse Maleka
Sifiso Nyambi

What went wrong? It was observed that Procedure ASDP001 point 2.3 states that checklist ASMF 6.3.1 will be used to actively monitor, measure and manage the performance of the company's suppliers. But at the time of assessment, there was no evidence that the labs external service providers were evaluated.

Why were the external service providers not evaluated as stated by ASP001?

> Personell failed to follow written procedures ASDP001 in the evaluation of external service providers.

Why were personnel not following the stated subclause in ASDP001

> Personnel used previous institutional engagements to assess the suitability of the external service providers & did not formally formalize the process of assessing & evaluate external service providers.

Why did personnel not formalize the process & implement it in assessing & evaluating the service providers.

> Oversight, non-adherence to procedures.

Supervisor: Maponya

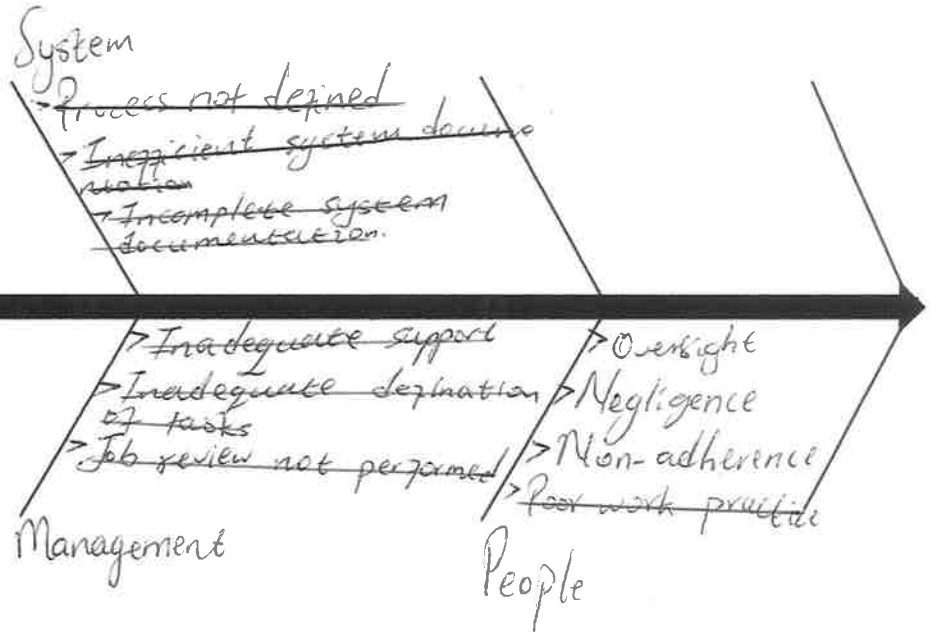
AS001MF

07/25

ROOT CAUSE ANALYSIS FORM

IRQ NO: AB23

At It was observed that procedure HSD P001 point 7.5 states that checklist ASMT 63.1 will be used to actively monitor, measure and manage the performance of company's supplier. But there was no evidence that the lab's external service providers were not evaluated.



SUPERVISOR

AzurESky "ensure quality of test"	Equipment Management		
	Document no: ASMF 6.3 Proc		
	Compiled by:	Tshepo Maponya - Laboratory Supervisor	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

1. Purpose & Scope

This procedure outlines the selection, identification, calibration, and maintenance of all laboratory equipment. This applies to all "critical equipment"—any hardware or software that can significantly influence the accuracy of laboratory results.

2. Selection and Verification

Before any equipment is placed into service (whether it is brand new or returning from repair), it must be verified.

- **Action:** The lab Manager/Supervisor compares the equipment's performance against the manufacturer's specifications or the laboratory's method requirements.
- **Record:** Identification & Verification results are noted in the **Equipment Register (ASMF 6.3 Equipment Register).**

3. Identification and Labelling

All equipment listed in the register shall be uniquely identified with a dedicated equipment number.

- **Calibration Labels:** Equipment must be labelled with its calibration status, indicating the **date of last calibration** and the **due date for the next calibration,** where necessary.

4. Handling, Transport, and Storage

To prevent contamination, damage, or deterioration:

- Equipment must be used only by authorized personnel.
- Sensitive instruments (e.g., balances) must be protected from vibrations and extreme temperature shifts.
- If equipment is transported outside the lab, it must be re-verified upon return before being used for testing.

Effective date	Rev status	Next review date
01/2026	00	01/2028

AFTER

	Equipment Management		
AzurESky	Document no: ASMF 6.4 Proc		
	Compiled by:	Tshepo Maponya - Laboratory Supervisor	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

1. Purpose & Scope

This procedure outlines the selection, identification, calibration, and maintenance of all laboratory equipment. This applies to all "critical equipment"—any hardware or software that can significantly influence the accuracy of laboratory results.

2. Selection and Verification

Before any equipment is placed into service (whether it is brand new or returning from repair), it must be verified.

- **Action:** The lab Manager/Supervisor compares the equipment's performance against the manufacturer's specifications or the laboratory's method requirements.
- **Record:** Identification & Verification results are noted in the **Equipment Register (ASMF 6.3 Equipment Register).**

3. Mandated Service Provider

To maintain technical consistency and metrological traceability, the Laboratory has appointed Micron Equip as the mandated provider for all equipment procurement, servicing, and calibration as detailed in the External Provider Checklist (Section 6.6.3). Software Management system has been mandated to QMWARE following their assessment & evaluation using checklist 6.6.3.

Disclaimer: Management reserves the right to evaluate a service provider at any time during the stipulated period of validity if critical issues arise with the agreement in place or a new and better service provider becomes available. The new service provider shall undergo the same process before approval.

3.1. Communication of Requirements

In accordance with ISO/IEC 17025 Clause 6.6.3, all technical and quality requirements are communicated to the provider via the signed checklist and supplementary purchase orders. This includes:

- **Technical Specifications:** Defined in the equipment table, including specific calibration setpoints and tolerances.
- **Acceptance Criteria:** Verification against the technical requirements of this procedure before equipment is commissioned.

- **Competence:** Mandatory requirement for **SANAS accreditation**, which in turn validates personnel competence as assigned technical signatories to their scope of activities listed in the accreditation certificate.
- **Access Rights:** The Laboratory reserves the right to witness activities at the provider's premises to verify compliance.

3.2. Ongoing Monitoring

The evaluator assesses the service provider annually; if no issues are found, the agreement with the service provider automatically rolls over to the next year. If issues arise during the review, a new service provider shall be evaluated the same way and approved for the services the lab requires when the criteria and competence requirements are met.

4. Identification and Labelling

All equipment listed in the register shall be uniquely identified with a dedicated equipment number.

- **Calibration Labels:** Equipment must be labelled with its calibration status, indicating the **date of last calibration** and the **due date for the next calibration,** where necessary.

4. Handling, Transport, and Storage

To prevent contamination, damage, or deterioration:

- Equipment must be used only by authorized personnel.
- Sensitive instruments (e.g., balances) must be protected from vibrations and extreme temperature shifts.
- If equipment is transported outside the lab, it must be re-verified upon return before being used for testing.

9. Revision

Revision no	Content changed	Reason for revision
01	• Added section 8, which controls the use of external equipment. • Added a section that covers the mandating of the equipment service provider (section 3)	Ensure compliance to clause 6.4.2 of ISO 17025 Ensures compliance to subclause 6.6.3 of ISO 17025

BEFORE

	Azuresky	
AzurESky	Management system requirements	Document no: ASDP001
	Compiled by: Sifiso Nyambi - Laboratory Manager	
	Approved by: Nonhlanhla Msibi - Operations Manager	

The assessment of data to determine the stability and homogeneity, assigned values and associated uncertainties of the properties or PT item, evaluate the performance of the PT participants and give opinions and interpretations as well as advice to the participants is performed by Laboratory Manager and Laboratory Supervisor.

The review and to authorise the information in the form of a PT report is the Operations Manager function.

2.2 Facilities and environmental conditions

The scope of the operation as defined as sample preparation of coal samples, analysis, packaging and storage. The available and allocated space of operation was planned and partitioned, demarcated to include sample preparation area, room housing furnaces, room housing CV machine and Inherent moisture oven and a storage room for finished products (PT samples and certified reference material).
Factors affecting the validity of results such temperature control etc, were defined and have been documented in procedure titled Environmental control with the registration number ASMF 7.2.3. This is to monitor, control and record the environmental conditions.

The minimum signage of safety requirements with regards to personal protective equipment were defined and affixed to the wall. This is as per the scope of operation.

2.3 Externally provided products and services

Azuresky (Pty) Ltd seeks to enhance the production of its activities and obtain externally provided products and services. As guided from ISO 17025:2017 clause 6.6 and 7.1, ISO 17034 clause 6.2 and ISO 17043 clause 6.4, it is the company's intention to incorporate externally provided products and services into laboratory's activities. They are intended to support the operation of the laboratory.

They include but not limited to the following products and services:
- Measurement standards eg Benzoic acid, etc
- Service and repair of equipment

BEFORE

	Azuresky		
AzurESky "ensure quality of test"	Management system requirements		Document no: ASDP001
	Compiled by:	Sifiso Nyambi - Laboratory Manager	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

- Mass and Temperature calibration of equipment
- Laboratory information management system
- Consumable materials, equipment, supply of equipment components
- Assessment and auditing services
- Manpower: a service level agreement has been entered into between Azuresky (Pty) Ltd and Hlogirati (Pty) Ltd for the provision of 4 (four) undergraduate students.

NB: The condition for acquiring the service of mass and temperature calibration of equipment and service and repair of equipment competent (not necessarily accredited) service providers are authorised to be used to provide the service.

BEFORE

The checklist titled "Provision of equipment, services and supplies" number ASMF6.3.1 was developed and shall be used to actively monitor, measure, and managing the performance of a company's supply's to ensure they meet the required quality, cost, delivery, and other performance expectations. As performance by suppliers can directly affect the overall efficiency, quality of products, and customer satisfaction of an organization. Furthermore, excessive slippage situations such as missed deadlines shall be flagged earlier than expected before things get out of hand allowing corrective measures to be taken more quickly! An early warning sheet ASMF 6.2 B is to be used to record such.

It's not all about the price. A company may receive invoices from suppliers following the agreed price, but unless a service or product meets expectations it is unlikely that the company can achieve savings target. It is a requirement that participants shall be informed, in advance and in writing, of products and services that are or can be provided externally, when they affect the validity of the PT activities.

Externally provided products and services procedure ASMF 6.6 has defined the purchasing requirements to be complied with. Document ASMF 6.3 is the policy required to be complied with for the provision of equipment, services and supplies.

AFTER

AzurESky "ensure quality of test"	Azuresky	
	Management system requirements	Document no: ASDP001
	Compiled by: Sifiso Nyambi - Laboratory Manager	
	Approved by: Nonhlanhla Msibi - Operations Manager	

The scope of operations includes **sample preparation of coal samples, testing and analysis, packaging, and storage**. The available and allocated operational space has been planned, partitioned, and demarcated to include:

- A sample preparation area
- A room housing furnace
- A room housing the CV machine and inherent moisture oven
- A storage room for finished products (PT samples and certified reference materials)

The **Laboratory Manager and Laboratory Supervisor** are responsible for all testing activities, including the preparation and evaluation of PT items and the production of reference materials. Their duties cover:

- Assessing data to determine stability, homogeneity, assigned values, and associated uncertainties of PT items and reference materials.
- Evaluating PT participant performance.
- Compiling reports and certificates for testing, PT activities, and reference materials.

The **Operations Manager** reviews and authorizes all compiled reports and certificates (for PT and CRMs only), ensuring formal validation before release. Factors affecting the validity of results—such as **temperature control and other environmental conditions**—have been defined and documented in the procedure *Environmental Control* (Registration No. ASMF 7.2.3). This procedure establishes monitoring, control, and recording requirements for environmental conditions. Minimum signage for safety requirements, particularly regarding personal protective equipment (PPE), has been defined and affixed to the walls in accordance with the scope of operations.

2.3 Externally provided products and services

Azuresky (Pty) Ltd seeks to enhance the production of its activities and obtain externally provided products and services. As guided from ISO 17025:2017 clause 6.6 and 7.1, ISO 17034 clause 6.2 and ISO 17043 clause 6.4, it is the company's intention to incorporate externally provided products and services into laboratory's activities. They are intended to support the operation of the laboratory.

They include but not limited to the following products and services:

- Measurement standards e.g., Benzoic acid, etc
- Service and repair of equipment
- Mass and Temperature calibration of equipment
- Laboratory information management system
- Proficiency Testing Scheme(s)
- Consumable materials, equipment, supply of equipment components
- Assessment and auditing services

AFTER

	Azuresky	
AzurESky "ensure quality of test"	Management system requirements	Document no: ASDP001
	Compiled by:	Sifiso Nyambi - Laboratory Manager
	Approved by:	Nonhlanhla Msibi - Operations Manager

- Manpower: a service level agreement has been entered into between Azuresky (Pty) Ltd and Hlogirati (Pty) Ltd for the provision of 4 (four) undergraduate students.

NB: The condition for acquiring the service of mass and temperature calibration of equipment, and service and repair of equipment, competent (not necessarily accredited) service providers are authorized to be used to provide the service.

The checklist(covered in equipment procedure 6.4) titled "Provision of equipment, services and supplies" number ASMF6.6.3 was developed and shall be used to actively select, monitor, measure, and manage the performance of a company's supplier to ensure they meet the requirements and acceptance criteria as set out in their respective checklist. When a supplier has been approved according to the criteria & requirements, the agreement is valid for a year. When it elapses, management shall evaluate the service received, and if it meets the stipulated requirements & no issues have arisen during the course of the year, the agreement rolls over to the following year. Upon re-evaluation, a section will be added on the checklist as proof of re-evaluation.

Disclaimer: Management reserves the right to evaluate a service provider at any time during the stipulated period of validity if critical issues arise with the agreement in place or a new and better service provider becomes available. The new service provider shall undergo the same process before approval.

AFTER

AzurESky "ensure quality of test"	Azuresky		
	Management system requirements		Document no: ASDP001
	Compiled by:	Sifiso Nyambi - Laboratory Manager	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

Revision

Revision no	Content changed	Reason for revision	Date changed
01	The manual has been rephrased to include testing services offered	The external audit revealed a gap within the system that needed to be addressed	15/04/2026
01	Include annual completion of declaration of interest document no. ASMF 4.2.1 by all personnel	Effective monitoring of the requirement	20/04/2026
01	Added a detailed section that explains how the evaluation of customer feedback is done, under 4.4.	To fully address customer feedback, Clause 8.6.2 of ISO 17025.	22/04/2026
01	Consolidated externally provided products and services to align with equipment procedure 6.4 & checklist 6.6.3	Alignment with the revised management system	22/05/2026

Disclaimer: The rev status for this document has been kept at 01 throughout the update of certain sections; this is to avoid discrepancies in evidence submitted for NCs in all areas where this quality manual is concerned. This only applies to the duration of closing SANAS NCs raised during the initial assessment.

Effective date	Rev status	Next review date
05/2026	01	05/2029

	PROVISION OF EQUIPMENT, SERVICES, AND SUPPLIES CHECKLIST		
AzurESky	Document no: ASMF6.6.3		
	Compiled by:	**Tshepo Maponya - Laboratory Supervisor**	
	Approved by:	**Nonhlanhla Msibi - Operations Manager**	

Supplier name: Micron Equip

Supplier contact: andile@micronequip.com

Product/service supplied:

The laboratory has designated Micron Equip as the exclusive vendor for the procurement, servicing, and calibration of the equipment outlined in the Table below:

Equipment	Description	Calibration Required?
25mm Crusher	25mm 3-phase jaw crusher, capable of crushing ±50mm coal down to size -25mm. The crusher must have adjustable jaws, have a guard mesh housing all mechanical components, and a ±100mm receiving mouth.	No, efficiency to be conducted by the lab
10mm Crusher	10mm 3-phase jaw crusher, capable of crushing ±25mm coal down to size ±25mm. The crusher must have adjustable jaws, have a guard mesh housing all mechanical components, and a ±50mm receiving mouth	No, efficiency to be conducted by the lab
3mm Crusher	3mm 3-phase jaw crusher, capable of crushing ±10mm coal down to size ±3mm. The crusher must have adjustable jaws, have a guard mesh housing all mechanical components, and a ±20mm receiving mouth.	No, efficiency to be conducted by the lab
Pulverisers	Pulverisers capable of milling coal from -3mm to -212µm. They must either use a cross-beat or hammers, possess a 212µm sieve with a receiver. Both 3-phase & 2-phase are acceptable.	No, efficiency to be conducted by the lab.
Homogenizer	A 3-phase homogenizer capable of processing a 210 L drum of pulverised coal.	No, efficiency done by the lab.
Oven	A 700L-1400L oven capable of reaching temperatures of 110-120 °C with ±2 heating fans.	Yes, at setpoints 40°C & 107°C

Effective date	Rev status	Next review date
05/2026	00	05/2029

AzurESky	**PROVISION OF EQUIPMENT, SERVICES, AND SUPPLIES CHECKLIST** **Document no:** **ASMF6.6.3**	
Compiled by:	**Tshepo Maponya - Laboratory Supervisor**	
Approved by:	**Nonhlanhla Msibi - Operations Manager**	

Oven	A 50L oven capable of reaching temperatures of ±110°C.	Yes, at setpoint 107°C
Furnace	A muffle furnace capable of reaching temperatures of up to ±815°C	Yes, at setpoints 500°C & 815°C, ramping profile.
Furnace	A muffle furnace capable of reaching temperatures of up to ±900°C.	Yes, at setpoint 900°C
Sieve Screens	- Sieve screens with a 450mm circumference and sizes in mm (50,40,25,22.4,10,6,4.75,3,1,0.5). - Sieve screen with ±100mm circumference and size 212µm. - 450mm circumference receiver. - ±100mm circumference receiver.	No, verification done by the lab.
Drycal Calorimeter System	- Drycal Single Vessel Calorimeter System - Single Vessel Drycal Calorimeter Complete with: 1 x Computer complete 1 x Calorimeter 1x Controller box 2x Bombs 1x Filling station 1x Cooler/ washer 1x Circulation Tank 1x UPS 1x Startup seal kit 1x Operation manual - Water Chiller System Complete - Drycal Oxygen Regulator Complete Fittings & Piping - Drycal Crucibles - Benzoic Acid Tablets, mass at ±0.5g, CV at 26.423 MJ/KG	No, calibration to be done by the lab using Benzoic acid tablets.

Effective date	Rev status	Next review date
05/2026	00	05/2029

CONTROLLED DOC

	PROVISION OF EQUIPMENT, SERVICES, AND SUPPLIES CHECKLIST Document no: ASMF6.6.3		
	Compiled by:	**Tshepo Maponya - Laboratory Supervisor**	
	Approved by:	**Nonhlanhla Msibi - Operations Manager**	

Analytical Balance	Capacity: 120g Readability: 0.1mg Pan Size: 80mm x 80mm	Yes
Top Pan Scale	Capacity: 8000g Readability: 0.1g Pan Size: 300mm x 210mm	Yes
Platform Scale	Capacity: 150kg Readability: 10g Pan Size: 400mm x 500mm	Yes
Mass Pieces	Class F1 Weights(100g,50g,20g,10g,5g,1g,0.5g) Mass Piece Cast Iron (20kg, 10kg, 5kg)	Yes
Cone & Quarter Cross	A cone & quarter steel cross at ±7m for both the legs	No
Desiccator	±20L desiccator, with moisture-absorbing silica gel desiccants.	No

Note: Consumables for any equipment purchased at Micron shall also be purchased at Micron Equip.

To be filled by the service provider

I, Andile Simelane on behalf of **Micron Equip** hereby mandate that **Micron Equip** will be able to undertake the Supply, Maintenance & Calibration of the aforementioned equipment.

Signature: 

Date: 14.05.2025

To Be Filled By Azuresky Pty Ltd

Competence & Acceptance Criteria

Effective date	Rev status	Next review date
05/2026	00	05/2029

CONTROLLED DOC

AzurESky	**PROVISION OF EQUIPMENT, SERVICES, AND SUPPLIES CHECKLIST** **Document no:** **ASMF6.6.3**	
Compiled by:	**Tshepo Maponya - Laboratory Supervisor**	
Approved by:	**Nonhlanhla Msibi - Operations Manager**	

The overall competence & acceptance criteria required for the purchase & calibration of equipment is a SANAS accreditation (mass & temperature metrology) & ± 5 years of supply, servicing & calibration of equipment.

Remarks: This appointment is based on Micron Equips' demonstrated technical competence and a proven track record of reliability established through previous institutional engagements. Consequently, Micron Equip is recognised as a SANAS-accredited, pre-approved, trusted supplier for all laboratory technical requirements.

Supplier approval

☒ **Approved**
☐ **Conditionally Approved (e.g., for low-risk items or pending documents)**
☐ **Not Approved**

Evaluator Name: Tshepo Maponya

Signature: 
Date: 15-05-2026

CONTROLLED DOC

Effective date	Rev status	Next review date
05/2026	00	05/2029

AzurESky	Approved External Service Provider Register		
	Document no:	Version:	Page:
	ASMF 6.6.3 Register	0	1
	Compiled by:	Tshepo Maponya - Laboratory Supervisor	
	Approved by:	Nonhlanhla Msibi - Operations Manager	

Company Name	Evaluation status (approved/not approved)	Comments
1. Micron laboratories - Calibration, repair & Maintenance service Procurement of equipment and consumables	Approved	- Accredited, scope of accreditation: Temperature and Mass metrology - Supplier has been in business, servicing clients for +20 years
2. QMWare Pty Ltd - LIMS rental service	Approved	- Competent - Supplier has been in business, servicing clients for +20 years (past experience from previous roles motivated the selection)
3. Linberts Calibration Services - Calibration of vernier caliper	Approved	- Accredited, scope of accreditation: Dimensional Metrology Supplier has been in business, servicing clients for +20 years

Effective date	Rev status	Next review date
05/2026	00	05/2029

AzurESky	ATTENDANCE REGISTER		
	Document no:	Version:	Page:
	ASMF 17043-8.4.1	0	1
	Compiled by:	Sifiso Nyambi - Laboratory Manager	
	Approved by:	Nonhlanhla Msibi - Operations Manager	
Date:			
Chair:		Scribe	
Subject:	Changes to 6.4 Equipment Procedure & ASDP001 Procedure Introduction of checklist 6.6.3.		
Absent:			

Present	Cell Number	Designation	Signature	Email address
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